Input system:
Dependent variables:x_1, x_3, x_2
Independent variables:t
Equations:

$$\frac{dx_1}{dt} = x_3^3 \sin\left(\cos\left(a\ln\left(x_2\right)^2 + bx_3\right)\right) + x_3^3 b \ln\left(x_2\right)^4 + \sin\left(a\ln\left(x_2\right)^2 + bx_3\right)^5 + \mathrm{Dv}\left(g^2, 1, (-1); x_1\right)$$

$$\frac{dx_2}{dt} = x_2^2 \cos\left(\sin\left(a\ln\left(x_2\right)^2 + bx_3\right)\right) + \cos\left(a\ln\left(x_2\right)^2 + bx_3\right)^4 + \mathrm{EK}\left(\sin\left(\sin\left(a\ln\left(x_2\right)^2 + bx_3\right)\right), x_1\right)$$

$$\frac{dx_3}{dt} = x_1 \mathrm{ch}\left(a\ln\left(x_2\right)^2 + bx_3\right)^2 + x_1 \sin\left(a\ln\left(x_2\right)^2 + bx_3\right) + \mathrm{sh}\left(a\ln\left(x_2\right)^2 + bx_3\right)^5$$

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Result:
Dependent variables:$s_9, s_{12}, s_7, x_3, s_2, s_6, s_{16}, s_8, s_{10}, s_1, x_1, s_{11}, s_{14}, s_{15}, x_2, s_{13}, s_4, s_5, s_3$
Independent variables:t
Additional variables:

$$s_1 = \ln\left(x_2\right)$$

$$s_2 = \mathrm{inv}\left(x_2\right)$$

$$s_3 = \cos\left(as_1^2 + bx_3\right)$$

$$s_4 = \sin\left(as_1^2 + bx_3\right)$$

$$s_5 = \sin\left(s_3\right)$$

$$s_6 = \cos\left(s_3\right)$$

$$s_7 = \mathrm{Dv}\left(g^2, 1, (-1); x_1\right)$$

$$s_8 = \mathrm{DV}\left(g^2, 1, (-1); x_1\right)$$

$$s_9 = \cos\left(s_4\right)$$

$$s_{10} = \sin\left(s_4\right)$$

$$s_{11} = \mathrm{EK}\left(s_{10}, x_1\right)$$

$$s_{12} = \mathrm{EKs}\left(s_{10}, x_1\right)$$

$$s_{13} = \mathrm{EKc}\left(s_{10}, x_1\right)$$

$$s_{14} = \mathrm{EKi}\left(s_{10}, x_1\right)$$

$$s_{15} = \mathrm{ch}\left(as_1^2 + bx_3\right)$$

$$s_{16} = \mathrm{sh}\left(as_1^2 + bx_3\right)$$

Equations:

$$\frac{dx_1}{dt} = x_3^3 bs_1^4 + s_4^5 + x_3^3 s_5 + s_7$$

$$\frac{dx_2}{dt} = s_3^4 + x_2^2 s_9 + s_{11}$$

$$\frac{dx_3}{dt} = s_{16}^5 + x_1 s_{15}^2 + x_1 s_4$$

$$\frac{ds_1}{dt} = s_2 s_3^4 + s_2 x_2^2 s_9 + s_2 s_{11}$$

$$\frac{ds_2}{dt} = (-1)\, s_2^2 s_3^4 + (-1)\, s_2^2 x_2^2 s_9 + (-1)\, s_2^2 s_{11}$$

$$\begin{aligned}\frac{ds_3}{dt} = {} & (-2)\, s_4 a s_1 s_2 s_3^4 + (-2)\, s_4 a s_1 s_2 x_2^2 s_9 + (-1)\, s_4 b s_{16}^5 \\ & + (-2)\, s_4 a s_1 s_2 s_{11} + (-1)\, s_4 b x_1 s_{15}^2 + (-1)\, s_4^2 b x_1\end{aligned}$$

$$\frac{ds_4}{dt} = 2s_3^5 a s_1 s_2 + 2s_3 a s_1 s_2 x_2^2 s_9 + s_3 b s_{16}^5 + 2s_3 a s_1 s_2 s_{11} + s_3 b x_1 s_{15}^2 + s_3 b x_1 s_4$$

$$\begin{aligned}\frac{ds_5}{dt} = {} & (-2)\, s_6 s_4 a s_1 s_2 s_3^4 + (-2)\, s_6 s_4 a s_1 s_2 x_2^2 s_9 + (-1)\, s_6 s_4 b s_{16}^5 \\ & + (-2)\, s_6 s_4 a s_1 s_2 s_{11} + (-1)\, s_6 s_4 b x_1 s_{15}^2 + (-1)\, s_6 s_4^2 b x_1\end{aligned}$$

$$\begin{aligned}\frac{ds_6}{dt} = {} & 2s_5 s_4 a s_1 s_2 s_3^4 + 2s_5 s_4 a s_1 s_2 x_2^2 s_9 + s_5 s_4 b s_{16}^5 \\ & + 2s_5 s_4 a s_1 s_2 s_{11} + s_5 s_4 b x_1 s_{15}^2 + s_5 s_4^2 b x_1\end{aligned}$$

$$\frac{ds_7}{dt} = s_8 x_3^3 bs_1^4 + s_8 s_4^5 + s_8 x_3^3 s_5 + s_8 s_7$$

$$\begin{aligned}\frac{ds_8}{dt} = {} & (-1)\, x_3^3 bs_1^4 s_7 p_1 x_1^2 + (-1)\, x_3^3 bs_1^4 s_7 p_2 x_1 + (-1)\, x_3^3 bs_1^4 s_7 p_3 + (-1)\, s_4^5 s_7 p_1 x_1^2 \\ & + (-1)\, x_3^3 s_5 s_7 p_1 x_1^2 + (-1)\, s_4^5 s_7 p_2 x_1 + (-1)\, x_3^3 s_5 s_7 p_2 x_1 + (-1)\, s_4^5 s_7 p_3 \\ & + (-1)\, x_3^3 s_5 s_7 p_3 + (-1)\, s_7^2 p_1 x_1^2 + (-1)\, s_7^2 p_2 x_1 + (-1)\, s_7^2 p_3\end{aligned}$$

$$\frac{ds_9}{dt} = (-2)\, s_{10}s_3^5 a s_1 s_2 + (-2)\, s_{10}s_3 a s_1 s_2 x_2^2 s_9 + (-1)\, s_{10}s_3 b s_{16}^5 + (-2)\, s_{10}s_3 a s_1 s_2 s_{11} + (-1)\, s_{10}s_3 b x_1 s_{15}^2 + (-1)\, s_{10}s_3 b x_1 s_4$$

$$\frac{ds_{10}}{dt} = 2s_9 s_3^5 a s_1 s_2 + 2s_9^2 s_3 a s_1 s_2 x_2^2 + s_9 s_3 b s_{16}^5 + 2s_9 s_3 a s_1 s_2 s_{11} + s_9 s_3 b x_1 s_{15}^2 + s_9 s_3 b x_1 s_4$$

$$\frac{ds_{11}}{dt} = 2s_{14}s_9 s_3^5 a s_1 s_2 + s_{12}s_{14}x_3^3 b s_1^4 + 2s_{14}s_9^2 s_3 a s_1 s_2 x_2^2 + s_{14}s_9 s_3 b s_{16}^5 + 2s_{14}s_9 s_3 a s_1 s_2 s_{11} + s_{14}s_9 s_3 b x_1 s_{15}^2 + s_{12}s_{14}s_4^5 + s_{14}s_9 s_3 b x_1 s_4 + s_{12}s_{14}x_3^3 s_5 + s_{12}s_{14}s_7$$

$$\frac{ds_{12}}{dt} = 2s_{13}s_{14}s_9 s_3^5 a s_1 s_2 + s_{12}s_{13}s_{14}x_3^3 b s_1^4 + 2s_{13}s_{14}s_9^2 s_3 a s_1 s_2 x_2^2 + s_{13}s_{14}s_9 s_3 b s_{16}^5 + 2s_{13}s_{14}s_9 s_3 a s_1 s_2 s_{11} + s_{13}s_{14}s_9 s_3 b x_1 s_{15}^2 + s_{12}s_{13}s_{14}s_4^5 + s_{13}s_{14}s_9 s_3 b x_1 s_4 + s_{12}s_{13}s_{14}x_3^3 s_5 + s_{12}s_{13}s_{14}s_7$$

$$\frac{ds_{13}}{dt} = (-2)\, s_{12}s_{14}s_9 s_3^5 a s_1 s_2 + (-1)\, s_{12}^2 s_{14}x_3^3 b s_1^4 + (-2)\, s_{12}s_{14}s_9^2 s_3 a s_1 s_2 x_2^2 + (-1)\, s_{12}s_{14}s_9 s_3 b s_{16}^5 + (-2)\, s_{12}s_{14}s_9 s_3 a s_1 s_2 s_{11} + (-1)\, s_{12}s_{14}s_9 s_3 b x_1 s_{15}^2 + (-1)\, s_{12}^2 s_{14}s_4^5 + (-1)\, s_{12}s_{14}s_9 s_3 b x_1 s_4 + (-1)\, s_{12}^2 s_{14}x_3^3 s_5 + (-1)\, s_{12}^2 s_{14}s_7$$

$$\frac{ds_{14}}{dt} = (-2)\, s_{10}s_{14}^3 s_{12}s_9 s_3^5 a s_1 s_2 + (-1)\, x_3^3 b s_1^4 s_{10}s_{12}^2 s_{14}^3 + (-2)\, s_{10}s_{14}^3 s_{12}s_9^2 s_3 a s_1 s_2 x_2^2 + (-1)\, s_{10}s_{14}^3 s_{12}s_9 s_3 b s_{16}^5 + (-2)\, s_{10}s_{14}^3 s_{12}s_9 s_3 a s_1 s_2 s_{11} + (-1)\, s_{10}s_{14}^3 s_{12}s_9 s_3 b x_1 s_{15}^2 + x_3^3 b s_1^4 s_{13}s_{14}^2 + (-1)\, s_4^5 s_{10}s_{12}^2 s_{14}^3 + (-1)\, s_{10}s_{14}^3 s_{12}s_9 s_3 b x_1 s_4 + (-1)\, x_3^3 s_5 s_{10}s_{12}^2 s_{14}^3 + s_4^5 s_{13}s_{14}^2 + x_3^3 s_5 s_{13}s_{14}^2 + (-1)\, s_7 s_{10}s_{12}^2 s_{14}^3 + s_7 s_{13}s_{14}^2$$

$$\frac{ds_{15}}{dt} = 2s_{16}a s_1 s_2 s_3^4 + 2s_{16}a s_1 s_2 x_2^2 s_9 + s_{16}^6 b + 2s_{16}a s_1 s_2 s_{11} + s_{16}b x_1 s_{15}^2 + s_{16}b x_1 s_4$$

$$\frac{ds_{16}}{dt} = 2s_{15}a s_1 s_2 s_3^4 + 2s_{15}a s_1 s_2 x_2^2 s_9 + s_{15}b s_{16}^5 + 2s_{15}a s_1 s_2 s_{11} + s_{15}^3 b x_1 + s_{15}b x_1 s_4$$